

EX.No. : 5

****SELECT with various clause – WHERE, pattern matching ****

AIM:

To view the records from the Saveetha Hospital table using SELECT commands with WHERE clause and pattern matching.

DESCRIPTION:

The SELECT statement is used to retrieve data from a table. The WHERE clause filters rows according to a condition. Pattern matching is performed using the LIKE operator. The % wildcard matches zero or more characters and the _ wildcard matches one character.

SELECT:

SELECT column1, column2, ... FROM table_name WHERE condition;

LIKE:

SELECT * FROM table_name WHERE column_name LIKE 'pattern';

In MySQL, % matches any string of zero or more characters and _ matches any single character.

Saveetha Hospital Dataset:

patients(patient_name, ward, treatment, doctor, operation, operation_fee)

Questions:

1. The hospital administrator wanted to display the patient name, ward and treatment for all patients.
2. Display all patients who have undergone an operation.
3. List the patients whose treatment is 'Cancer'.
4. Display all patients whose operation fee is greater than 10000.
5. Display all patient details whose doctor is not 'Paracetamol'.
6. List the patients whose name ends with the substring 'sh'.
7. Display all patients whose name contains the substring 'sh'.
8. Find all patients whose doctor name contains 'par' as a substring.
9. Display the patients whose names do not contain exactly six letters.
10. Find all the patients whose names contain 'am'.

OUTPUTS:

1. The hospital administrator wanted to display the patient name, ward and treatment for all patients.

```
mysql> select patient_name,ward,treatment from patients;
```

```
+-----+-----+-----+
| patient_name | ward | treatment |
+-----+-----+-----+
| Ajay         | 534  | Heart Attack |
| Tharun       | 213  | Accident    |
| Aishu        | 413  | BP          |
| Arun         | 321  | Heart       |
| Raja         | 247  | Tyroid      |
| Harish       | 412  | Cold        |
| Vadivu       | 631  | Fever       |
| Ashitha      | 443  | Lung Cancer |
| Kamini       | 881  | Cancer      |
| Balaji       | 921  | Stomach Pain |
+-----+-----+-----+
10 row(s) in set (0.00 sec)
```

```
mysql>
```

2. Display all patients who have undergone an operation.

```
mysql> select * from patients where operation='Yes';
```

```
+-----+-----+-----+-----+-----+-----+
| patient_name | ward | treatment | doctor | operation | operation_fee |
+-----+-----+-----+-----+-----+-----+
| Ajay         | 534  | Heart Attack | Astromeg | Yes | 100000 |
| Tharun       | 213  | Accident    | Nylor   | Yes | 100000 |
| Raja         | 247  | Tyroid      | Tyroid  | Yes | 10000  |
| Ashitha      | 443  | Lung Cancer | Amino   | Yes | 50000  |
| Kamini       | 881  | Cancer      | Dels    | Yes | 100000 |
+-----+-----+-----+-----+-----+-----+
5 row(s) in set (0.00 sec)
```

```
mysql>
```

3. List the patients whose treatment is 'Cancer'.

```
mysql> select * from patients where treatment='Cancer';
```

```
+-----+-----+-----+-----+-----+-----+
| patient_name | ward | treatment | doctor | operation | operation_fee |
+-----+-----+-----+-----+-----+-----+
| Kamini       | 881  | Cancer    | Dels    | Yes | 100000 |
+-----+-----+-----+-----+-----+-----+
1 row(s) in set (0.00 sec)
```

```
mysql>
```

4. Display all patients whose operation fee is greater than 10000.

```
mysql> select * from patients where operation_fee > 10000;

+-----+-----+-----+-----+-----+-----+
| patient_name | ward | treatment | doctor | operation | operation_fee |
+-----+-----+-----+-----+-----+-----+
| Ajay        | 534  | Heart Attack | Astromeg | Yes | 100000 |
| Tharun      | 213  | Accident    | Nylor   | Yes | 100000 |
| Ashitha     | 443  | Lung Cancer | Amino   | Yes | 50000  |
| Kamini      | 881  | Cancer      | Dels    | Yes | 100000 |
+-----+-----+-----+-----+-----+-----+
4 row(s) in set (0.00 sec)

mysql>
```

5. Display all patient details whose doctor is not 'Paracetamol'.

```
mysql> select * from patients where doctor not like 'Paracetamol';

+-----+-----+-----+-----+-----+-----+
| patient_name | ward | treatment | doctor | operation | operation_fee |
+-----+-----+-----+-----+-----+-----+
| Ajay        | 534  | Heart Attack | Astromeg | Yes | 100000 |
| Tharun      | 213  | Accident    | Nylor   | Yes | 100000 |
| Aishu       | 413  | BP          | BP Tablet | No | 5000   |
| Arun        | 321  | Heart       | Borevita | No | 10000  |
| Raja        | 247  | Thyroid     | Tyroid   | Yes | 10000  |
| Ashitha     | 443  | Lung Cancer | Amino   | Yes | 50000  |
| Kamini      | 881  | Cancer      | Dels    | Yes | 100000 |
+-----+-----+-----+-----+-----+-----+
7 row(s) in set (0.00 sec)

mysql>
```

6. List the patients whose name ends with the substring 'sh'.

```
mysql> select * from patients where patient_name like '%sh';

+-----+-----+-----+-----+-----+-----+
| patient_name | ward | treatment | doctor | operation | operation_fee |
+-----+-----+-----+-----+-----+-----+
| Harish       | 412  | Cold      | Paracetamol | No | 1000   |
+-----+-----+-----+-----+-----+-----+
1 row(s) in set (0.00 sec)

mysql>
```

7. Display all patients whose name contains the substring 'sh'.

```
mysql> select * from patients where patient_name like '%sh%';

+-----+-----+-----+-----+-----+-----+
| patient_name | ward | treatment | doctor | operation | operation_fee |
+-----+-----+-----+-----+-----+-----+
| Aishu       | 413  | BP          | BP Tablet | No | 5000   |
| Harish      | 412  | Cold        | Paracetamol | No | 1000   |
| Ashitha     | 443  | Lung Cancer | Amino   | Yes | 50000  |
+-----+-----+-----+-----+-----+-----+
3 row(s) in set (0.00 sec)

mysql>
```

8. Find all patients whose doctor name contains 'par' as a substring.

```
mysql> select * from patients where doctor like '%par%';
```

patient_name	ward	treatment	doctor	operation	operation_fee
Harish	412	Cold	Paracetamol	No	1000
Vadivu	631	Fever	Paracetamol	No	1000
Balaji	921	Stomach Pain	Paracetamol	No	1000

```
3 row(s) in set (0.00 sec)
```

```
mysql>
```

9. Display the patients whose names do not contain exactly six letters.

```
mysql> select * from patients where patient_name not like '_____';
```

patient_name	ward	treatment	doctor	operation	operation_fee
Ajay	534	Heart Attack	Astromeg	Yes	100000
Aishu	413	BP	BP Tablet	No	5000
Arun	321	Heart	Borevita	No	10000
Raja	247	Tyroid	Tyroid	Yes	10000
Ashitha	443	Lung Cancer	Amino	Yes	50000

```
5 row(s) in set (0.00 sec)
```

```
mysql>
```

10. Find all the patients whose names contain 'am'.

```
mysql> select * from patients where patient_name like '%am%';
```

patient_name	ward	treatment	doctor	operation	operation_fee
Kamini	881	Cancer	Dels	Yes	100000

```
1 row(s) in set (0.00 sec)
```

```
mysql>
```

RESULT:

The records from the Saveetha Hospital table are successfully displayed using SELECT commands with WHERE clause and pattern matching.